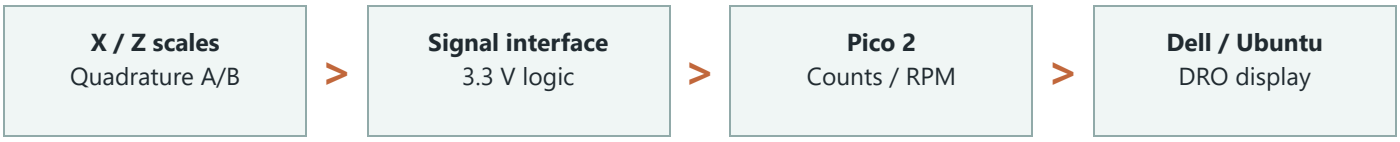
INSTALLATION, WIRING AND OPERATOR GUIDE

Lathe Digital Readout

Raspberry Pi Pico 2 + Dell Ubuntu display

Wiring the X/Z scales and PINDA spindle pickup, flashing firmware, starting the DRO, and checking readings before use.



Electrical safety: Pico GPIO is not 5 V tolerant. Your PINDA has a reported 5 V signal output: never wire that output directly to GP6. This firmware provides no input protection or level conversion. Use a suitable 5-to-3.3 V signal interface before the Pico, and verify its output is safe.

System settings in this project

Function	Firmware/application assumption
X and Z scales	X GP2/3; Z GP4/5; quadrature transitions
Spindle pickup	GP6 falling-edge timing; 4 pulses per spindle revolution assumed
Linear conversion	25,400 decoded counts per inch by default; verify against actual travel
Host connection	USB serial; packet contains X count, Z count, and RPM

This is a measurement and calculation aid, not a motion controller or safety interlock. It does not stop or control the lathe. Continue using guards, the machine manual, and independent measuring tools.

Identify the sensor interface first

Power down before wiring. Stop the spindle, isolate machine power as appropriate, unplug the Pico, prevent accidental movement, and keep wiring clear of rotating parts, full axis travel, chips, coolant, and mains/VFD wiring.

- Every GPIO input must stay between 0 V and 3.3 V. 5 V, 12 V, 24 V, or negative-going signals can damage the Pico.
- Scale inputs GP2-GP5 have no firmware pull-up/down. Many industrial scales are differential (such as RS-422); use the correct receiver. Never connect A-/B- directly to GPIO.
- GP6 has an internal pull-up and detects falling edges. The internal pull-up is to 3.3 V; it is not a 5 V level shifter or protection device.
- Your PINDA's signal output is reported as 5 V. Connect it to GP6 only through a suitable 5-to-3.3 V interface whose input matches the PINDA's output type and whose Pico-side output stays within 0-3.3 V.
- Verify the exact PINDA connector pinout, supply, ground, and output type from documentation; do not identify wires by color alone. Power the probe only as specified, and do not use the Pico 3V3 pin unless the probe documentation and current requirements explicitly allow it.
- For non-isolated interfaces, connect a common signal reference only as the interface documentation specifies. Maintain isolation where the interface calls for it. Do not assume the lathe frame or cable shield is signal ground.

PINDA as a spindle pickup

1. Confirm the exact probe model, connector pinout, output type, and sensing range from its documentation.
2. Because this probe's signal is 5 V, install the matching 5-to-3.3 V signal interface between the probe and GP6. Verify the interface output voltage before connecting the Pico. Do not rely on the Pico's internal pull-up to reduce a 5 V signal.
3. Choose a secure spindle target that this probe detects at the installed gap. Do not use a loose target near rotating machinery; ensure mounting cannot detach or strike anything.
4. Confirm the conditioned output changes state once per target and produces a clean falling edge at GP6.
5. Count targets per revolution. Firmware currently assumes four falling edges per revolution. If your target setup differs, the firmware constant must be changed to the actual number.
6. Confirm the probe's maximum switching rate supports the target count at intended RPM. Validate against a trusted tachometer with guards in place.

Safe first power-up: Flash firmware with all sensor signals disconnected. Verify the USB link, then add one correctly interfaced signal at a time with power removed between wiring changes.

Signal wiring plan

The table shows the firmware GPIO and corresponding Pico 2 header pin. Use the GP labels and the official Pico 2 pinout to confirm board orientation before wiring.

Conditioned interface output	Pico GPIO	Physical pin	Use
X scale A	**GP2**	4	X quadrature A
X scale B	**GP3**	5	X quadrature B
Z scale A	**GP4**	6	Z quadrature A
Z scale B	**GP5**	7	Z quadrature B
PINDA tach logic output (verified 0-3.3 V)	**GP6**	9	Falling-edge input
Interface output ground, if required	**GND**	3 or 8	Signal reference for compatible non-isolated interfaces

```
X interface OUT-A -------------------------> Pico pin 4  (GP2)
X interface OUT-B -------------------------> Pico pin 5  (GP3)
Z interface OUT-A -------------------------> Pico pin 6  (GP4)
Z interface OUT-B -------------------------> Pico pin 7  (GP5)
PINDA interface OUT (verified safe logic) --> Pico pin 9  (GP6)
Interface output GND (as specified) --------> Pico GND pin 3 or 8

Probe/scale power: use its specified external supply/interface.
Do not connect its supply voltage to Pico GPIO.
```

Signal-level connection map only; it is not a power-supply schematic. Fit the correct differential receiver, isolator, pull-up, or level shifter based on actual device documentation.

Assembly and routing

- Use insulated connectors and strain relief suitable for machine vibration, chips, and coolant.
- Route each signal pair with its return or as the interface specifies. Separate low-level signals from VFD, motor, contactor, and mains wiring.
- Secure cables and check full carriage/cross-slide travel with power isolated so nothing pulls, pinches, or fouls.
- If an axis direction is reversed, power down and swap that axis's A and B logic outputs at the interface, then retest.
- Verify one falling tach edge per intended target. Do not add arbitrary capacitors or pull-ups; they may distort pulses or conflict with the interface.

Flash and start the Pico 2

Flash the supplied UF2 on Windows

1. Stop the DRO on the Dell. Unplug the Pico from the Dell and use a USB data cable on Windows.
2. Hold BOOTSEL while plugging the Pico into the PC. Release when detected. Device Manager may show **RP2350 Boot**; the boot volume commonly appears as **RPI-RP2**.
3. Copy `build-pico2/lathe_dro_pico.uf2` from the project folder onto the Pico boot volume. Do not format or initialize it.
4. Wait for the volume to disappear as the Pico restarts. Reconnect it to the Dell normally, without BOOTSEL.

Optional firmware rebuild on Ubuntu

The project uses CMake, Ninja, ARM GCC, and Pico SDK 2.3.1.

```
sudo apt update
sudo apt install cmake ninja-build gcc-arm-none-eabi git python3
git clone --branch 2.3.1 --recurse-submodules \
  https://github.com/raspberrypi/pico-sdk.git pico-sdk
git -C pico-sdk submodule update --init --recursive
chmod +x build_pico2.sh
./build_pico2.sh
```

Output: `build-pico2/lathe_dro_pico.uf2`. Building does not flash the Pico; use BOOTSEL.

Dell Ubuntu application

A graphical desktop/session is required for Pygame; headless SSH alone cannot display the app.

```
sudo apt install python3 python3-venv python3-pip
cd /path/to/lathe-dro-project
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
sudo usermod -aG dialout "$USER"
# Log out/in after changing group membership
chmod +x run_dro.sh
./run_dro.sh
```

The app auto-detects a Pico USB serial port. If necessary, specify the actual port, e.g. `./run_dro.sh --port /dev/ttyACM0`. To stop a foreground DRO, press `Ctrl+C` in its terminal. Restart after reconnecting/reflashing.

Optional demo: `./run_dro.sh --demo --windowed`. Simulated readings are not for calibration or machining.

> **Serial format:** firmware sends lines like `X:123,Z:-456,RPM:300` at about 60 updates/second. Raw counts restart at zero after Pico reboot; establish the work reference again.

Verify before machining

X and Z scale calibration

The app defaults to **25,400 decoded counts per inch**. Scale vendors may define cycles differently; SETUP calibration measures an effective scale factor but does not prove the scale's stated accuracy.

1. Stop the spindle. Select INCH/MM on the main screen before entering calibration distance.
2. Measure a long known straight-axis travel with a trusted standard. Approach the start consistently and account for backlash.
3. In **SETUP**, select the X or Z distance field, enter the known travel, position at the start, and click **CAPTURE X START** or **CAPTURE Z START**.
4. Move that axis by the entered distance, then click **CALIBRATE X** or **CALIBRATE Z**. The factor uses raw count change, independent of zero and display modes.
5. For X, enter actual cross-slide (radial) movement, not twice that amount from the DIA display. Recheck over a second distance and in both directions when safe.
6. Use the +/- 0.0001 buttons for fine adjustment only after checking mounting, missed counts, backlash, and signal integrity.

> **Persistence:** calibrated X/Z factors and fine adjustments are saved in `scale_calibration.json` beside the app. Factors correct the base 25,400-counts/in assumption; verify against a trusted standard before machining.

Readout controls

Control	Meaning
ZERO X / ZERO Z	Set origin for the selected axis and mode.
ABS / INC	Absolute or incremental display; entering INC starts a fresh incremental origin.
RAD / DIA	X radius or diameter; diameter display is twice radial travel.
INCH / MM	Display units and calibration-entry units.
T-1 ... T-9	Select tool ID. Offsets apply in ABS when both T-1 and selected tool have touch-offs.
TOOL SET	Capture T-1 master at a datum first, then each tool at the same datum. Stored in `tool_offsets.json` beside the app.
FEED / SPEED / TAPER / COORD	Reference calculators only; they do not operate the lathe. Verify chart and setup values independently.
SETUP	Calibrate X/Z from known travel, fine-adjust factors, and change display resolution. Factors persist in `scale_calibration.json`.
POWER	Requests Ubuntu shutdown after confirmation; does not stop the lathe.

Boring estimate

Enter the measured starting-hole diameter, position the tool at the starting bore wall, and capture X. The app adds twice the calibrated radial X travel. Choose X direction so the estimate increases while boring outward. Verify final diameter with a bore gauge.

RPM verification

Firmware measures falling-edge intervals; current constant is four pulses/revolution. Confirm actual target count and PINDA switching rate at intended spindle speed. Compare with a trusted tachometer using guarding and safe procedure. RPM is advisory and does not control the spindle.

Commissioning checks

Symptom	Checks
Pico not found	Reconnect in normal mode; use a data USB cable; check Linux serial permissions/device path; restart DRO after flashing.
Scale does not count	Power down. Check device/interface supply, safe logic level, signal reference, A/B wiring, connectors, and GP2/3 or GP4/5.
Axis direction reversed	Power off and swap that axis's A/B interface outputs; retest known travel.
Erratic counts	Check receiver type, voltage, return path, loose terminals, electrical noise/cable routing, shield termination, and scale alignment. Do not guess pull-ups/capacitors.
RPM stays zero	Check exact PINDA version, target material/gap, interface output, active-low GP6 signal, and secure target mounting.
RPM wrong by ratio	Count falling edges per revolution; firmware assumes four. Ensure the probe can switch fast enough.
Readings drift	Compare several distances; check calibration, missed counts, backlash, scale mounting, and mechanical alignment.

Before first use

[ ] Exact scale and PINDA part numbers and output specifications recorded.
[ ] Required level shifting/receivers/isolation installed and checked.
[ ] All Pico GPIO signals verified between 0 and 3.3 V.
[ ] Wiring insulated, secured, labeled, and clear through full travel.
[ ] X/Z direction and travel checked against trusted standards in multiple positions.
[ ] RPM checked against a trusted tachometer with guards in place.
[ ] Operator confirms ABS/INC, RAD/DIA, unit, and tool selection before measuring.

Stop using the DRO if readings jump, reverse unexpectedly, disagree with trusted measurements, or wiring is damaged. Isolate power and correct the cause before machining. This system is not a safety interlock or motion controller.

Project-specific assumptions match this repository's current firmware/application. PINDA generations and scale interfaces vary. Verify hardware documentation and the lathe's own speed chart for the installed equipment.